Now, we shall find the GCD of $x^3 - 5x^2 + 10x - 8$ and $x^3 - 4x^2 + 7x - 6$

$$
\begin{array}{r|l}
 & 1 \\
\hline
x^3 - 5x^2 + 10x - 8 & x^3 - 4x^2 + 7x - 6 \\
 & x^3 - 5x^2 + 10x - 8 \\
 & (-)\ \ (+)\ \ (-)\ \ (+) \\
\hline
 & x^2 - 3x + 2
\end{array}
$$

$$
\begin{array}{r|l}
 & x - 2 \\
\hline
x^2 - 3x + 2 & x^3 - 5x^2 + 10x - 8 \\
 & x^3 - 3x^2 + 2x \\
 & (-)\ \ (+)\ \ (-) \\
\hline
 & -2x^2 + 8x - 8 \\
 & -2x^2 + 6x - 4 \\
 & (+)\ \ (-)\ \ (+) \\
\hline
 & 2x - 4
\end{array}
$$

$$= 2(x - 2)$$

$$
\begin{array}{r|l}
 & x - 1 \\
\hline
x - 2 & x^2 - 3x + 2 \\
 & x^2 - 2x \\
 & (-)\ \ (+) \\
\hline
 & -x + 2 \\
 & -x + 2 \\
 & (+)\ \ (-) \\
\hline
 & 0
\end{array}
$$

Here, we get zero remainder.

GCD of leading coefficients 3 and 6 is 3.

Thus, GCD $\left[(6x^3 - 30x^2 + 60x - 48,\ 3x^3 - 12x^2 + 21x - 18)\right] = 3(x - 2)$.

3.3.2 Least Common Multiple (LCM) of Polynomials

The Least Common Multiple of two or more algebraic expressions is the expression of highest degree (or power) such that the expressions exactly divide it.

Consider the following simple expressions $a^3b^2,\ a^2b^3$.

For these expressions $LCM = a^3b^3$.

To find LCM by factorization method

(i) Each expression is first resolved into its factors.

(ii) The highest power of the factors will be the LCM.

(iii) If the expressions have numerical coefficients, find their LCM.

(iv) The product of the LCM of factors and coefficient is the required LCM.

Example 3.12 Find the LCM of the following

(i) $8x^4y^2,\ 48x^2y^4$ (ii) $5x - 10,\ 5x^2 - 20$

(iii) $x^4 - 1,\ x^2 - 2x + 1$ (iv) $x^3 - 27,\ (x - 3)^2,\ x^2 - 9$

10th_Maths_Chapter 3_English.indd 95 12/23/2024 5:32:42 PM